

# Response to Reviewers

## Paper 1217: Depth, Breadth, and Bias: Structural Diffusion of Political Content on Divergent Platforms

### Summary

We thank the SPC and the three reviewers for their careful and constructive comments. We were encouraged that the reviewers found the question timely and relevant to ICWSM (SPC, R2, R3) and the cross-platform comparison valuable, particularly because Truth Social and Bluesky remain understudied in comparative research (SPC, R1). In the meanwhile, we found the feedback very helpful in identifying several places where the original manuscript needed clearer definitions, additional validation, and more cautious interpretation, and we have revised the manuscript substantially in response.

Across the sensitivity analyses described below, estimates sometimes change substantially in magnitude, but none overturns the central conclusions: that reposts scale alike on the two platforms, that replies do not, and that most of the reply difference is associated with who participates and how they engage across ideological lines. Where one of them narrows the interpretation of a particular finding, we have added that qualification to the manuscript.

Below we respond first to the seven priorities in the SPC’s roadmap and then to each reviewer’s comments in turn. Reviewer comments are shown in italics and our responses in blue. Where a reviewer’s comment is addressed previously, we summarize the change and refer to that response.

### Responses to the SPC

- ***SPC-1.** Validate or revise the repost cascade analysis. Provide robustness checks showing how results change under alternative repost reconstruction assumptions, including assumptions about follower-network timing and inferred exposure paths. If this cannot be done convincingly, consider reducing the emphasis on repost cascades or focusing the main contribution on reply cascades.*
  - Thank you for this suggestion, which has made the paper considerably more robust. We added a new appendix (Appendix D, pp. 17-18) that rebuilds all 244,129 repost cascades (4.03 million reposts) under alternative reconstruction rules. Reconstructing a repost cascade means deciding, for each repost, which earlier post the user saw it through. The possible sources are the original post and any earlier repost by an account the user follows. The rule used in the paper attaches a repost to the original post whenever the user follows the original author, and otherwise to an earlier reposter the user follows. The alternatives pick the earliest possible source, the most recent one, or one chosen at random, a draw we repeated 40 times. Comparing attribution rules in this way follows Bakshy et al. (2011), who faced the same problem on Twitter and reported their results

under several such rules; Appendix D now cites them. The quantity we compare is the platform difference in scaling: how much more steeply breadth (the most posts at any one level of a cascade) or depth (how many levels of replies it has) rises with cascade size on Truth Social than on Bluesky. It is the platform-by-size interaction in the paper’s regressions, which relate the logarithm of breadth or depth to the logarithm of size, so it is a difference in slopes. For reply cascades, it is  $+0.170$  for breadth and  $-0.182$  for depth. For repost cascades, the lowest and highest values across all the rules are  $-0.012$  and  $+0.031$  for breadth and  $-0.083$  and  $+0.028$  for depth (Figure A4, p. 18). Both ranges span zero, since some rules give a slightly negative difference and others a slightly positive one, and even the largest repost value is less than a fifth of the reply value for breadth and less than half of it for depth. The rule used in the paper gives the largest depth difference of any rule, and for breadth every rule stays within 0.031 of zero. The choice of rule also has limited room to matter: 55 percent of Bluesky reposts and 59 percent of Truth Social reposts have only the original post as a possible source. The conclusion that reposts scale similarly on the two platforms therefore holds under every rule we examined.

- On follower-network timing: the follower networks carry no dates for when each follow was created, so a follow edge could postdate the repost it is used to explain. On both platforms, however, the follower lists were collected together with the posts and reposts, one day apart, so they reflect the same period as the reposts and any such error applies to both platforms alike. As a check, we used the rule that attaches each repost to the most recent possible source and deleted 5 percent of follow edges at random before reconstruction, likely more than such timing error could produce. The estimates barely move: the platform difference is 0.028 for breadth and 0.005 for depth, against 0.031 and 0.004 with no deletion, and stays far below the reply-cascade values of  $+0.170$  and  $-0.182$  (Appendix D, pp. 17-18).
- We also removed the description of repost cascades as empirical. They are called reconstructed throughout, and Section 3.2 (pp. 4-5) states how they are built and when the follower lists were collected.
- **SPC-2.** *Clarify cascade definitions. Explicitly state whether nodes are users, posts, or interactions; how repeated users in the same thread are represented; and how these choices affect depth, breadth, and motif analyses.*
  - Thank you for raising this! Each node is one action by a user: the root post, a reply, or a repost. A user who replies several times in one thread therefore appears once per reply, and each node carries the ideology label of the account that wrote it. The submitted version described the nodes as users, which holds in the sense that every node is a user’s action, but it was misleading because it did not say whether repeated appearances by the same user are merged. Section 3.2 (p. 4) now states that they are not, for both cascade types, and why. A reply tree records how a conversation actually unfolded, and merging a user’s posts would add a modeling assumption to that record. For example, if user a posts, b replies, a answers b, and c then replies to a’s answer, c’s reply sits three levels below the root; merging a’s two posts would place it one level below the root, as if c had answered a’s opening post directly. Depth, breadth, and the motif counts are therefore computed on the threads as they occurred.
  - The motif analysis (Figures 4 and 5) counts small recurring patterns of three connected posts, such as a chain of three replies, and it does so on the post-level trees, so one user can occupy two positions in a motif. To show how often that happens, we added

a new appendix (Appendix E, p. 18) that counts exchanges of the form  $A \rightarrow B \rightarrow A$ , in which a reply’s author also wrote its grandparent post. Among cascades of depth two or more, 68.8 percent on Bluesky and 70.0 percent on Truth Social contain at least one, and such exchanges account for 20.0 and 15.6 percent of reply edges. Exchanges of this kind are similarly common on both platforms. Threads between only two accounts are more common on Bluesky, but dropping them leaves most of the depth difference in place (see our response to Reviewer 1 on two-user exchanges).

- **SPC-3.** *Strengthen ideology-label validation. Report more detailed validation results, including class-specific and platform-specific performance. Clarify the meaning of the center category and test whether the main findings are robust to plausible ideology-label noise.*

- Thank you for these suggestions, which, we believe, substantially strengthened the measurement side of the paper. We added new sections to Appendix B. The first reports class-specific and platform-specific precision and recall, with bootstrap intervals, against the 171 of 200 validation items on which both human annotators agree (Table A2, p. 17). Accuracy against this consensus is 0.86. On all 200 items, agreement between the model and each annotator ( $\kappa = 0.61$  and  $0.74$ ) is largely comparable to agreement between the two annotators ( $\kappa = 0.77$ ). The errors are not uniform: the model over-assigns each platform’s ideological minority, with precision of 0.64 on both platforms for the minority label (right-leaning on Bluesky, left-leaning on Truth Social), against 0.98 and 0.93 for each platform’s largest class. Because the minority cells rest on few items, we test by bootstrap whether the gap is real (one-sided  $p = 0.017$  on Bluesky and  $p = 0.004$  on Truth Social) and report its size with its uncertainty.
- The center category is now defined explicitly as a residual. A user is labeled left- or right-leaning when more than 60 percent of their posts lean that way, and the center category collects everyone else: users whose posts alternate between left and right, and users who consistently post without an overt ideological stance (Section 3.5, p. 5). We interpret center-labeled users that way throughout, including in the motif analysis. How many users fall in the center category depends on the 60 percent threshold, so we re-estimated the models with thresholds of 0.5 and 0.7 (Appendix B, Figure A3 and Table A1, pp. 16-17). Center-labeled users remain at every threshold, including 0.5, and their share grows as the threshold rises. Composition and alignment account for 63 percent of the platform difference in breadth and 67 percent in depth at 0.5, 84 and 86 percent at 0.6, and 98 and 96 percent at 0.7. The share rises with the threshold because a stricter threshold places more users in the center category, which composition then separates from the partisan groups.
- We also added a label-noise analysis (Appendix B, pp. 15-17). It asks how much of the platform difference composition and alignment would still account for if the labels contained errors at the rates we measured. With the original labels, on the 120,693 cascades used in this analysis, they account for 84 percent of the platform difference in breadth and 86 percent in depth. In each simulation draw, every user’s label is redrawn from the measured confusion matrix, composition and alignment are recomputed, and the regression with both ideology measures is refit. The baseline model, which has no ideology measures, does not use the labels, so its platform difference is the same in every draw; label noise can change only how much of that difference the two measures account for. We ran 100 draws under each of four error processes: the measured error rates for each platform; one set of error rates pooled across both platforms; error rates re-estimated from a resampled validation sample in every draw, which carries the uncertainty of the small

validation sample into the simulation; and the less favorable rates measured against a single annotator. In the median draw of each process, the share accounted for is between 70 and 86 percent, and in the least favorable of all 400 draws it is 66 percent (Table A3, p. 17). The pooled process lowers it the most; the platform-specific and resampled processes, which are closest to the error we measured, leave it near the original values. Label noise therefore lowers the share under some assumptions, but composition and alignment account for a majority of the platform difference in every simulation.

- **SPC-4.** *Reframe explanatory claims. Replace causal language suggesting that ideology and alignment “explain” platform differences with more cautious language about association, unless additional analyses can support a stronger claim. Discuss the potential endogeneity between alignment ratio and cascade structure.*
  - We are grateful for this comment and agree. We have revised the manuscript throughout: composition and alignment are now described as associated with, or statistically accounting for, the platform difference in the abstract, introduction, results, and discussion. Where the submitted version said that platform differences disappear once ideology is modeled, the revision reports how much of the difference composition and alignment account for: roughly five sixths of the platform difference for both breadth and depth (Table 2, p. 8).
  - On endogeneity: interaction alignment is computed from the same replies that determine depth and breadth, so alignment and cascade structure are jointly realized features of one conversation. Cross-ideological exchanges may lengthen a discussion, and the shape of a discussion may in turn change how many such exchanges it can hold: as R3 notes, a deep chain of replies and a star-shaped cascade, in which many users reply to one post, offer different openings for them. The Limitations section (Section 6, p. 10) states this and treats alignment as associated with cascade structure, and the Results point to it where the model with both ideology measures is introduced (p. 8).
- **SPC-5.** *Improve reporting of data and robustness. Add readable descriptive statistics on posts, cascades, cascade sizes, motif frequencies, user participation patterns, and topic model outputs. Include motif counts and explain how dependence among motifs is handled.*
  - Thank you for these suggestions and we think those details make the paper more comprehensive. Section 4.1 now includes a new Table 1 (p. 7), which reports, for each platform and cascade type, the number of cascades and posts, the share of cascades that are a single post, and the share larger than 50 posts, and the text interprets it. The full size distribution, as counts with percentages in four size ranges, is in a new Table A4 (Appendix C, p. 18). The data cover May 30 to June 30, 2024 (Section 3.1, p. 4). Compiling these tables surfaced something we had not reported: most reply cascades are a single post with no replies, and far more so on Bluesky (71.3 percent) than on Truth Social (43.3 percent), while 9.0 percent of Truth Social reply cascades exceed 50 posts, against 0.3 percent on Bluesky. Repost cascades are more alike in size. These single posts are identical in structure but not in frequency. Because every single post has the same size, breadth, and depth, they all fall on one point in the regression of breadth or depth on size, and the number of them on each platform pulls on that platform’s fitted line. We therefore refit every model on the 47,670 cascades with more than one post and report the results in a new Table A6 (Appendix H, p. 19). The platform difference is smaller on this sample, and composition and alignment still account for 81 to 92 percent of it under the two estimators Table A6 compares.

- We added a new appendix (Appendix F, pp. 18-19) whose Table A5 (p. 19) gives the observed count of each of the 54 ideology-labeled three-node motifs on both platforms, next to the z-scores in Figures 4 and 5, which compare each observed count with its average in randomized versions of the same trees, described below. On dependence: we count every three-node chain in a tree, including chains that share nodes, so a chain of four posts contains two overlapping three-node chains and both are counted. The randomized trees are counted the same way, so overlap raises the observed and the expected counts alike and does not bias the comparison. Overlap does make the scores of neighboring motifs correlated, so we read them together, as patterns across groups of related motifs, such as all chains whose three users share one label. Appendix F also describes the null model: 100 randomized trees per platform, built by swapping which posts replies attach to within each cascade, which keeps each cascade’s size, each post’s number of replies, and each participant’s ideology fixed and randomizes who replies to whom. The submitted main text gave the number of randomizations as 1,000,000, which did not match the code or the appendix; both now state 100.
- User participation patterns, in particular how often the same two users reply to each other, are reported in the new Appendix E (see SPC-2). A new Appendix G (p. 19) reports the topic model: it was fit on 125,623 root posts, the clustering step first sets aside posts that fit no topic well, BERTopic’s outlier reduction step then assigns all but 21 of them (0.02 percent of posts) to their nearest topic, and manual labeling yields the eleven categories we use as topic controls in the regressions.
- The SPC’s summary also asked about back-and-forth reply chains between the same two users and about figure readability; we address these below, in our responses to Reviewer 1’s comments on two-user exchanges and on figures.
- **SPC-6.** *Resolve methodological inconsistencies. Ensure that the regression estimator, loss function, and standard error procedure are described consistently across the main text and appendix.*
  - Thank you for catching this, and we apologize for the confusion; it was an error in the submitted version. The reply-cascade models are robust linear models with the Huber loss, which limits how much any single cascade with an unusual shape can pull the fit. The appendix paragraph describing ordinary linear regression with HC3 standard errors, a common correction for unequal error variance, was left over from an earlier specification and has been removed. The only analysis estimated by OLS with HC3 standard errors is the new repost reconstruction check, where the breadth and depth of small repost trees take only a few distinct values, so the Huber scale estimate collapses to zero; the appendix now says so.
  - Checking this surfaced a related problem. Because 61 percent of reply cascades on the two platforms combined are a single post, and all single posts have the same size, breadth, and depth, the default median-absolute-deviation scale estimate collapses to nearly zero, and the standard errors in the submitted Table 1 could not be interpreted. We refit every model with Huber’s proposal 2 scale estimate, a standard alternative that estimates the spread of the residuals jointly with the coefficients and does not collapse here, and report standard errors from the sandwich covariance estimator, the standard choice for robust regressions of this kind (Section 3.6, p. 5; Table 2, p. 8). Point estimates shift. The largest change is in the platform difference for breadth in the model with both ideology measures, which moves from 0.023 to 0.071; the same difference is 0.419 in the baseline model without ideology measures, and composition and alignment still account

for about five sixths of the platform difference for both outcomes. A new Table A6 (Appendix H, p. 19) reports the results under both scale estimators, with and without single-post cascades: the share accounted for ranges from 81 to 94 percent across all four combinations of sample and estimator. The appendix analyses of alternative ideology thresholds, an alternative measure of composition based on each platform’s majority and minority ideology, and label noise use the same estimator as Table 2.

- **SPC-7.** *Sharpen implications and limitations. Explain why the observed structural differences matter, what they suggest about platformed political discussion, and how limited the conclusions are to the 2024 U.S. election context and Biden/Trump-related discourse.*
  - Thank you for this suggestion! The Discussion (Section 5, pp. 9-10) now states what the two cascade shapes mean. Wide, shallow reply cascades consist of many users responding to the same post with little exchange among them; narrow, deep cascades consist of replies that themselves draw replies over several turns. The two shapes distribute attention differently, and in our data the difference appears even though both platforms offer the same kind of threaded replies. The Discussion also states a methodological implication: comparisons based on activity or cascade size alone would record Truth Social as simply more active and miss the divergence, which appears only after conditioning on size.
  - The limitations are now a separate section (Section 6, p. 10). It states three limits. The sample covers conversations mentioning Biden or Trump during one month of the 2024 campaign, so the ideological mix we observe describes those conversations and should not be read as either platform’s full user base, and we cannot tell whether the same patterns hold for other topics. The ideology labels are coarse positions within these conversations. And the alignment result is an association. The conclusion the evidence supports is correspondingly narrower: in conversations about U.S. politics, platforms with similar reply mechanics can exhibit different conversational structures conditional on cascade size, and that these differences are associated with who participates and how they engage across ideological lines.

## Responses to Reviewer 1

### Repost cascade reconstruction

- *You have reconstructed the cascades for reposts in a very deterministic manner according to section 3.2. I have no way of gauging whether this reconstruction is in any way realistic and I am concerned that the results which show that Bsky and Truth Social have similar report cascade structure are an artefact of this reconstruction process. I would want to see substantial validation and robustness checks to assess the influence of your choice of reconstruction mechanism on your results. The text refers to “empirical” cascades - if your cascades are reconstructed, then these are not empirical but modelled. I suggest either (1) cutting all the repost content and focusing on the reply trees, or (2) adding substantial robustness checks which assess the impact of the cascade reconstruction mechanism.*
  - Thank you for this detailed comment and for laying out the two options! We took the second. Because the concern is that one deterministic rule could produce the similarity, we re-derived every repost cascade under alternative deterministic rules, as Bakshy et al. (2011) did for Twitter, and under a random-rule ensemble, and found that the rule used in the paper gives the largest depth difference of any of them. Under all of them

the platform difference for reposts stays close to zero, far below that for reply cascades (new Appendix D and Figure A4, pp. 17-18; details under SPC-1). We also removed “empirical” for these cascades, which are now called reconstructed throughout.

## Ideology labels

- *A lot of the results rely on a comparison between different ideological groups (left, centre, right). This is interesting, but it is also extremely reliant on the method used to label the ideology of users. This concerns me for multiple reasons: (1) you seem to find vastly more left leaning users of Truth Social than the wider literature would suggest, and likewise far more right leaning users on Bsky than the literature would suggest (I am thinking of a Bsky paper by Alex Bovet’s team in Zurich which looked into this). This makes me question your ideological labelling and all the results which depend on this labelling, e.g., the ideology-based cascade trees. It is also important because the literature consistently suggests that most social media users are not politically engaged (although this is of course less likely to be the case in threads mentioning Biden or Trump). I would ask the authors to go to some effort to convincing the reader of the validity of their ideology labelling - currently I am not convinced.*
  - Thank you for pressing us on this, and for pointing to the Bluesky study by Bovet’s group (Quelle and Bovet 2025), which Appendix B now compares with our estimates (p. 17). Our estimates describe users who take part in the political conversations our keywords define, posts mentioning Biden or Trump during one campaign month, and they are not estimates for either platform’s full user base. As you note, most users are not politically engaged, and less political users rarely appear in these conversations, so the ideological mix we observe need not match platform-wide estimates. Our labels also summarize the stance each user expresses in these conversations, which can differ from that user’s broader political position. The Limitations section now states both points (p. 10).
  - Part of the gap is also measurement error, in the direction you suspected: the model over-assigns each platform’s ideological minority (SPC-3). We added a correction for this to Appendix B (p. 17), which adjusts the estimated shares using the measured confusion matrix. It lowers the right-leaning share on Bluesky from 24.2 to 17.9 percent and the left-leaning share on Truth Social from 11.5 to 1.9 percent. Quelle and Bovet classify 4.8 percent of Bluesky users as right of center, and Nogara et al. (2026) 7.2 percent, so a gap remains for Bluesky after correction; it may reflect our sample of posts mentioning Biden or Trump and differences in measurement across studies.
  - Since our sample is not the platform population, the relevant test is whether the labels are reliable enough for the comparison they are used in. The new analyses under SPC-3 address this directly: class-specific and platform-specific validation, a test of the direction of the error, simulated label noise at the measured rates, and alternative thresholds. Across all of them, composition and alignment account for a majority of the platform difference.

## Motifs, topics, and data

- *I like Figures 4 and 5, but I would want to know how many of each motif are actually present in your dataset. This is crucial information for actually assessing the validity of your results.*
  - Thank you for this suggestion. A new Table A5 (Appendix F, p. 19) gives the observed

count of all 54 ideology-labeled motifs on both platforms, alongside the z-scores in Figures 4 and 5. The counts matter for interpretation in the way you anticipated: on Truth Social, for example, reply chains that go back and forth between a left-leaning and a right-leaning user (left, right, left, or the reverse) are strongly overrepresented relative to the randomized trees while remaining rare in absolute terms, which the Discussion now notes (p. 9).

- *Related to point 2 - different motifs are of course not independent of each other. Does your randomisation strategy account for this. In a chain of length 4, there are two 3 motifs, but these are not independent. How does this effect your results?*
  - Thank you for this question. The randomization does account for it, in the sense that matters for the z-scores. We count every three-node chain, including chains that share nodes, in both the observed and the randomized trees. A chain of four posts therefore adds two overlapping three-node chains to both counts, and the overlap does not bias the comparison. It does make the scores of neighboring motifs correlated, so we read them together, as patterns across groups of related motifs, and say so in Appendix F (pp. 18-19; see also SPC-5).
- *A bit more info on the topic modelling would be useful. In my experience most Twitter-like posts are too short to get a useful topic label and BERTopic tends to throw 50% + of the posts into an outlier group.*
  - We appreciate this suggestion. A new Appendix G (p. 19) describes the topic model. It was fit on 125,623 root posts. The clustering step inside BERTopic first sets aside posts that fit no topic well; we applied BERTopic’s outlier reduction step with its default settings, which assigned all but 21 of them (0.02 percent of posts) to their nearest topic. Manual labeling of the resulting clusters yields the eleven categories we use as topic controls in the regressions. Topic accounts for little of the platform difference, so the uncertainty that reassignment introduces is unlikely to affect the main comparison.
- *I want far more info about the data. How many posts? Over how long? The CCDFs in Fig 2 are useful to a point, but I cannot read off how many cascades of size N there actually are. Can you provide more of this data in a readable format?*
  - A new Table 1 in Section 4.1 (p. 7) reports, for each platform and cascade type, the number of cascades and posts and the share of cascades that are a single post or exceed 50 posts, and Table A4 (Appendix C, p. 18) gives the number of cascades in each size range with percentages. The data cover May 30 to June 30, 2024 (Section 3.1, p. 4). There are, for example, 79,348 reply cascades on Bluesky containing 199,413 posts, and 43,841 on Truth Social containing 1,369,497 posts. The table also led to an additional analysis excluding single-post cascades, described under SPC-5.

## Further analyses

- *I may have missed this, but how many of the reply chains are conversations between two users, i.e., a back and forth between the same two accounts?*
  - Thank you for this question, which we had not examined before. We added this analysis to the paper as a new appendix (Appendix E, p. 18). We measured it two ways. Counting whole threads: among threads in which some reply draws a further reply, 28.4 percent on Bluesky and 6.5 percent on Truth Social involve only two accounts. Truth Social threads are much larger, with a median of 18 participants against 3 on Bluesky, so

two-person threads are rarer there. Counting exchanges: for every reply we checked whether its author also wrote the post two levels above it, an  $A \rightarrow B \rightarrow A$  exchange. Such exchanges occur in 68.8 percent of these threads on Bluesky and 70.0 percent on Truth Social, and make up 20.0 and 15.6 percent of reply edges. To test whether two-person threads drive the depth difference, we refit the models without them. The depth difference is  $-0.184$ , against  $-0.222$  among all threads with a reply, and composition and alignment still account for a majority of it. Two-person conversations are more common on Bluesky, but they do not explain the depth difference.

- *I'll be honest, I don't fully follow the results about the alignment of users... Some clearer explanation would be useful.*
  - Thank you for this feedback! The section was not as clear as it should have been. We rewrote the definition in Section 3.5 (p. 5) and added a worked example. Each reply edge is coded aligned if both users share an ideological label and misaligned otherwise, and a cascade's alignment ratio is the share of aligned edges. In the example, a right-leaning user A writes the root post. Two right-leaning users, B and C, reply to A; a left-leaning user D also replies to A; and another left-leaning user E replies to D. The replies from B, C, and E are aligned and the reply from D is misaligned, so the alignment ratio is  $3/4 = 0.75$ . The measure describes the conversation: the same user can contribute aligned edges in one thread and misaligned edges in another.
- *The justification for only looking at influencers on TS is poor. The lack of a natural gap in the cascade distribution is not really a natural justification. What if you just take the top n% of accounts on both platforms?*
  - Thank you for pressing on this and for suggesting a concrete alternative! The original manuscript did not explain clearly why we made this choice, and we now state the reasoning explicitly in Appendix A (p. 14) and summarize it where H1 is stated (Section 2.4, pp. 3-4). It lies in the shape of the two follower distributions. On Truth Social, eight accounts are separated from the rest of the distribution: the largest has 56 times the followers of the platform's 99.9th-percentile account, and together the eight hold 16.4 percent of all followers in our data. On Bluesky the largest account sits five times above the 99.9th percentile, continuous with the rest of the distribution, and the top eight hold 6.1 percent. The eighth-largest Truth Social account has more followers than the largest Bluesky account.
  - A top-n percent rule would therefore not equalize the comparison: on Bluesky it would remove accounts continuous with the rest of the distribution, and on Truth Social any n large enough to capture the separated accounts would also sweep in ordinary ones. The regression that tests for influencers (Model 2 in Table 2, p. 8) accordingly treats the eight Truth Social accounts as a separate group. Separating them changes the platform difference only slightly, from 0.419 to 0.412 for breadth and from  $-0.321$  to  $-0.313$  for depth.
- *I don't particularly like that this is another US centric paper, but that is not really reason to reject. Just a shame not to diversify the focus.*
  - Thank you for raising this point. We agree with the concern, and we also wish the empirical focus of social media research were less concentrated on the U.S. context. At the same time, the U.S. focus in this paper follows from the case itself. The 2024 U.S. presidential election was an unusually high-salience and dramatic political moment, and Truth Social and Bluesky provide a particularly useful comparison because their political

identities and user communities are closely connected to that political context. We see this as an empirically valuable setting for studying candidate-centered political diffusion, while also recognizing that it is not a basis for broad cross-national generalization. We have revised the Limitations section accordingly to make the U.S.-centered scope explicit and to note the need for future comparative work across national contexts.

## Figures, presentation, and citations

- *A lot of the figures need a bit of cleaning up / further explanation.*
  - Thank you for this feedback. We revised the figures. In the motif figures (Figures 4 and 5, pp. 9-10), the labels naming each group of motifs had been drawn inside the panels, on top of the data. They now sit in a band above the axes. The captions explain why scores with  $|Z| < 3$  are grayed out, and Figure 4’s caption explains the groupings. In the topic and partisanship distributions (Figures 1 and A2, pp. 6 and 15), value labels on the longest bars fell outside the axes; both figures are rebuilt so every label sits inside, and the partisanship figure now uses a shared axis. The figure that checks whether each ideology measure needs a curved term in the model (Figure A6, p. 22) was three separate images. It is now one row of panels, each showing a fitted curve beside the straight-line fit, so the curvature the appendix describes is visible. All figures are rendered at 300 dpi.
- *Overall the presentation is good, but there is a fair bit of unexplained notation. Also, the figures are messy in places with labels overlapping parts of the content. It is not always clear what is being shown. The latex has paragraph indents in places where there shouldn't be any.*
  - Thank you for pointing these out. We reviewed the manuscript for undefined notation and define the terms of the statistical models where they first appear (Section 3.6, p. 5). The stray paragraph indents came from blank lines after displayed equations and have been removed. The figure changes are described in our response to your first general comment.
- *The citations are appropriate and comprehensive. However, the literature is quite old in the context of social media research. It would perhaps be worth updating the literature to some papers which are more recent on information spreading on social media.*
  - Thank you for this suggestion. We added recent work on information spreading. It includes DeVerna et al. (2024) on how the cascade inference problem distorts measured diffusion, González-Bailón et al. (2024) on the diffusion and reach of information on Facebook during the 2020 U.S. election, and Slaughter et al. (2025) on how Community Notes reduce the diffusion of false information. We also added recent methodological work on using automated and language-model labels in downstream analysis (Egami et al. 2023; TeBlunthuis, Hase, and Chan 2024; Vallejo Vera and Driggers 2025). The Related Work section now cites 17 studies published since 2023, including work on Bluesky and Truth Social.

## Responses to Reviewer 2

### Causal interpretation

- *I feel that the main “ideology explains the platform gap” claim is too strong: The paper measures ideological composition and alignment from the same completed reply cascades whose*

*depth and breadth it is trying to explain. That makes these variables partly endogenous to the outcome. Alignment is not clearly a prior cause of cascade shape; it may simply describe the cascade after it has already formed. I think the safer claim is that ideology and alignment are associated with the platform gap, not that they explain it causally.*

- Thank you for this comment; we fully agree. Throughout the paper, composition and alignment are now described as associated with the platform difference. The Limitations section (Section 6, p. 10) discusses the endogeneity you describe: alignment and cascade structure are produced by the same conversation, so their association does not show which influences which. Details under SPC-4.

## Cascade construction

- *The cascade construction is ambiguous for me. The paper says cascade nodes are users, but reply trees are naturally built from posts/replies. If the same user replies multiple times, collapsing posts into one user node can change depth, breadth, and motif structure. Can authors further clarify this?*
  - Thank you for pointing this out; the text was unclear on this point. Nodes are posts: a user who replies several times contributes one node per reply, each carrying that user’s label (Section 3.2, p. 4). We keep these appearances separate because a reply tree records how a conversation actually unfolded: collapsing them into one user node would move a user’s later replies, and the replies those draw, to positions where they did not occur, adding a modeling assumption to observed data. Details under SPC-2.
- *In addition, the paper infers repost paths because platform APIs do not directly provide true diffusion paths. That is reasonable, but the inference depends on assumptions about follower relationships and timing. This may be further justified.*
  - Thank you for this suggestion! This is a common concern raised by all reviewers. We now test these assumptions directly. All repost cascades are rebuilt under alternative linking rules and under simulated follower-timing error, and the repost result holds throughout (Appendix D, pp. 17-18). Details under SPC-1.

## Generalizability

- *The study covers one month and only Biden/Trump-related discussion during a highly unusual campaign period, which makes the generalizability a little limited.*
  - Thank you for this comment. We agree that this limits what the paper can claim. The new Limitations section (Section 6, p. 10) states that the sample consists of posts mentioning Biden or Trump during one month of the 2024 campaign, and it limits the conclusion to conversations about U.S. politics. Details under SPC-7.

## Ideology validation

- *Another concern is about data size. The LLM-based stance validation uses only 200 human-labeled replies, and the paper does not report enough class-specific or platform-specific performance. The “center” category also appears to include genuinely centrist users, mixed users, and uncertain users. Given ideology is the paper’s main explanatory variable, label noise could directly affect the results.*

- Thank you for these suggestions. We now report class-specific and platform-specific performance with bootstrap intervals (Table A2, p. 17), define the center category explicitly as a residual and show how its size and the results change with the classification threshold, and propagate the measured label error through the models. The minority-class cells in the 200-item sample are small, so we report their uncertainty, test the direction of the error by bootstrap, and show that the substantive result survives label noise at the measured rates. Details under SPC-3.

## Regression estimator

- *In addition to the weaknesses listed above, I find that the methods describe robust regression with Huber loss, while the appendix says linear regression with HC3 robust standard errors. These are different estimators. I suggest authors double check this.*
  - Thank you for catching this, and we apologize for the inconsistency! The models use the Huber loss, and the HC3 paragraph has been removed. Checking this also led us to correct the scale estimator and refit the regression table (now Table 2). Details under SPC-6.

## Responses to Reviewer 3

### Treatment of ideological alignment

- *The main claim that ideological composition and interaction alignment jointly account for the platform-level divergence in reply cascade scaling is interesting, but this interpretation requires caution. The alignment ratio is computed from the reply edges that constitute the cascades. Therefore, while alignment ratio may affect cascade structure, it is also possible that cascade structure affects the alignment ratio. For example, deep cascades and star-like cascades may provide different opportunities for cross-ideological interaction. I therefore recommend weakening the causal interpretation of this result, or strengthening the discussion and analysis of this issue.*
  - Thank you for this comment and for the example of deep and star-like cascades, which helped us state the problem clearly. We weakened the causal interpretation throughout, and the Limitations section now states that the relationship can run in both directions (Section 6, p. 10). Your example shows how: a deep chain gives opposing users repeated turns to answer one another, while a star-shaped cascade mostly gathers separate replies to one post, so the shape of a conversation changes how much cross-ideological exchange it can hold. Details under SPC-4.

### Construction of repost cascades

- *The paper uses the follower network to infer repost cascades, but the timing of post collection and follower-network collection may affect the accuracy of this inference. For example, suppose user  $u$  follows user  $v$  and  $u$  reposts  $v$ 's post. The paper appears to infer that information flowed from  $v$  to  $u$ . However, is it possible that  $u$  followed  $v$  only after reposting  $v$ 's post? If  $u$  followed  $v$  after the repost occurred, then inferring information flow from  $v$  to  $u$  would not be appropriate. The paper should discuss the potential impact of this type of inference error. It should also consider whether such errors may differ systematically between Truth Social and Bluesky.*

- Thank you for this careful example. It describes a real limitation: the follower networks carry no dates for when each follow was created, so a follow edge can postdate the repost it is used to explain. On both platforms, however, the follower lists were collected together with the posts and reposts, one day apart, so they reflect the same period as the reposts, and the error applies to both platforms alike, which leaves it little room to differ systematically between them. As a check, we deleted 5 percent of follow edges at random before reconstruction, likely more than such timing error could produce; the platform difference for reposts barely moves (0.028 for breadth and 0.005 for depth, against 0.031 and 0.004 with no deletion) and stays far below the reply-cascade values of +0.170 and -0.182 (Appendix D, pp. 17-18). Section 3.2 (pp. 4-5) now states when the follower lists were collected. Details under SPC-1.

### Definition of cascades

- *In reply cascades, the same user may appear multiple times in the same thread. It is unclear how the paper handles such cases. For example, does the paper allow the same user to appear as multiple nodes in a single cascade? Or are nodes defined strictly as users, with only a specific reply, such as the first or last appearance, counted for each user? These details should be clarified.*
  - Thank you for raising this. The same user can appear as multiple nodes: nodes are posts, and a user who replies several times contributes one node per reply (Section 3.2, p. 4). We keep these appearances separate because a reply tree records how the conversation actually unfolded, and merging them would add a modeling assumption to observed data. Details under SPC-2.

### Implications and limitations

- *The finding that cascade structures differ across platforms is interesting. However, the practical and theoretical implications of this finding are not very clear. It is also unclear whether the results are specific to the 2024 U.S. election context or whether they can be generalized more broadly. The generalizability of the findings may be addressed in future work, but the paper should provide a more careful discussion of its implications and limitations.*
  - Thank you for this suggestion. We rewrote the Discussion to state what the two cascade shapes imply (Section 5, pp. 9-10) and moved the limitations, including the dependence on the 2024 U.S. campaign and on a sample of posts mentioning Biden or Trump, into a separate section (Section 6, p. 10). Details under SPC-7.

We thank the reviewers and the SPC again for their careful and constructive feedback. We feel that this revision has made the project substantially stronger and more robust. The comments pushed us to clarify several parts of the analysis, add sensitivity checks that we had not originally included, and narrow some of the interpretations where appropriate. We very much appreciate the time the reviewers spent on the paper and the opportunity to revise it.